Building Web and Mobile Apps WS2025/26 (Prof. Dr. Cathrin Möller)

Exercise 0 - 20./21.10.2025

Disclaimer: The exercise is intended to be done manually - without Chat GPT. If you have any questions, feel free to work in a group or ask tutors or me for help. We will ensure at the end of the practical exercise class (/ at the beginning of next week) that you are able to solve the tasks.

What you will need to earn the up to 3 points per week for the portfolio will always be highlighted in class. Contrary to „Distributed Applications“, points earned in this class will count into your grade.

As basic web technologies like HTML and CSS build the foundation to more complex frameworks, it should be a learning goal to understand what higher level frameworks are doing internally. Therefore we start with a repetition using HTML and CSS from scratch.

Theory reflection part 1

T1) What is the purpose of HTML and CSS in a website? Why are those technologies often separated in files?

T2) How can I add CSS to HTML? (3 Options)

Practical Exercises part 1

P1) Create a „bwma“ (name) project folder with a „vanilla“ sub folder and file „index.html“ within the vanilla folder. Create a GitLab (https://git-ce.rwth-aachen.de/) repo to push your code into. **Initialize a private GitLab repo „bwma-25-26“** and invite ***@cathrin.moeller*** as a „**Reporter**“. You will need this in the following weeks for the submissions. It also helps as we will continue working on the same application or look up things already done for multiple weeks. You can add members via Manage => Members => Invite members.

Commit the empty vanilla application to your GitLab repo running „***git init***“ in the root folder of your app. „***git add ***“ will add all code. „***git commit -m "0-Create vanilla“***“ will add a commit with a proper message. Follow the instructions in GitLab to push to your repo.

Hint: In case of problems, there is a „force push“ option in branch rules, which can be activated temporarily.

P2) One of the most complicated tasks in CSS is proper centering. Create a section within your HTML file and a layout with a blue box that is only 300px wide. It should be centered horizontally.

P3) Write „Hello World“ inside the blue box and ensure it is **centered** horizontally and vertically inside the box.

P4) In this **index.html**, now create a new **section** with a layout containing **two boxes** side by side, each 100px high, the left one is blue, the right one is red. They should stretch each over 50% of the page width. Provide a „from scratch“ CSS solution applying **display: inline-block** to both boxes. Do you observe something unexpected? How can we solve this problem?

Commit the solution to the repo.

P5) Modern web technologies try to simplify centering using different **display** properties like flex and grid.

If you are not familiar yet using flex and grid, you can learn it with games ;-)

Let’s deep dive into flexbox and grid layouting by playing

Flexbox Froggy **https://flexboxfroggy.com**

or Grid Garden **https://cssgridgarden.com**

Create another section in the HTML containing two green boxes. Provide an alternative solution using flex or grid and commit and push to the repo.

P6) **Responsive design** is of high importance nowadays. Create a new section that has a fluid layout depending on screen width. On screen widths of mobile phones we should see 2 boxes side by side in one row, on tablet screens 3 and on desktop browsers 4.

Hint: use @media queries in css and a combination of grid and flex

Commit and push to the repo.

Bonus: Theory reflection part 2

T3) Consider the following HTML. Create some CSS that ensures the text in each element is shown with a different colour. What is the specificity of each selector?

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>My first exercise</title>
</head>
<body>
<div id="container">
  <section class="content">
    <h1>Welcome to Specificity Exercises</h1>
    <p class="text">This is a paragraph with
      <span class="highlight">highlighted</span>
      text.</p>
    <p id="unique">An element with a unique identifier.</p>
    <ul>
      <li class="item">Item 1</li>
      <li>Item 2</li>
      <li class="special item">Special Item</li>
    </ul>
  </section>
  <a href="#" class="link">More exercises</a>
</div>
</body>
</html>
```

T4) Style the "Special Item" list element to have a blue background colour by creating a CSS selector with a specificity score of 0,2,1,1.